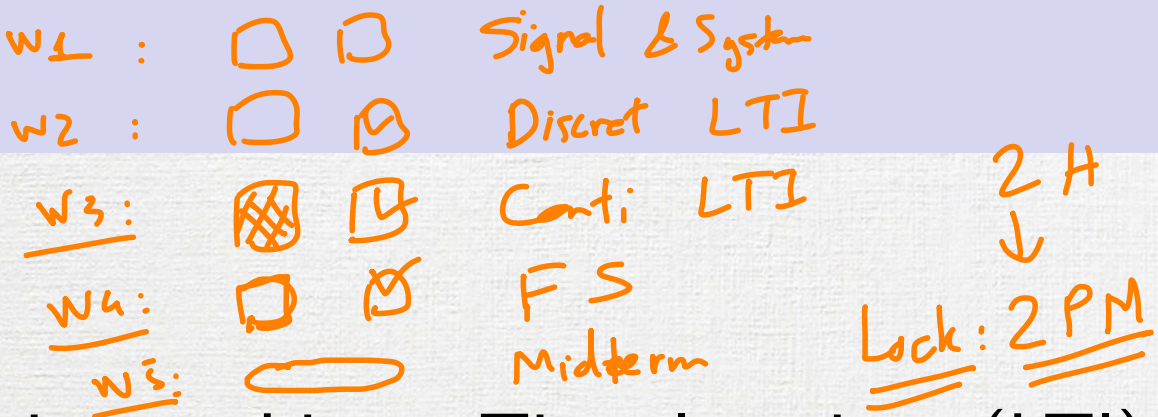


Continuous Linear Time-Invariant (LTI) Systems

TRAN Hoang Tung
LE Nhu Chu Hiep

Information and Communication Technology (ICT) Department
University of Science and Technology of Hanoi (USTH)

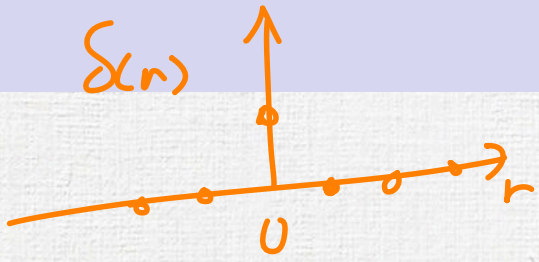
October 01, 2025

Objectives

Continuous Delta

Convolution Integral

Homework



1 Lesson Objectives

2 Continuous-time Delta

3 Convolution Integral

4 Homework

δ(t) | δ(n)

∫ | Σ

1 Lesson Objectives

2 Continuous-time Delta

3 Convolution Integral

4 Homework

Lesson Objectives

Signals & Systems

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Objectives

Continuous Delta

Convolution Integral

Homework

At the end of this lesson, you should be able to

Lesson Objectives

Signals & Systems

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Objectives

Continuous Delta

Convolution Integral

Homework

At the end of this lesson, you should be able to

1. understand the origin of the Delta signal in continuous-time domain

Lesson Objectives

At the end of this lesson, you should be able to

1. understand the origin of the Delta signal in continuous-time domain
2. calculate convolution integral

1 Lesson Objectives

2 Continuous-time Delta
- Weighted "Sum" of Shifted Impulses
- Definition
- Discrete vs Continuous

3 Convolution Integral

4 Homework

Review: Sum vs Integral

Signals & Systems

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Objectives

Continuous Delta

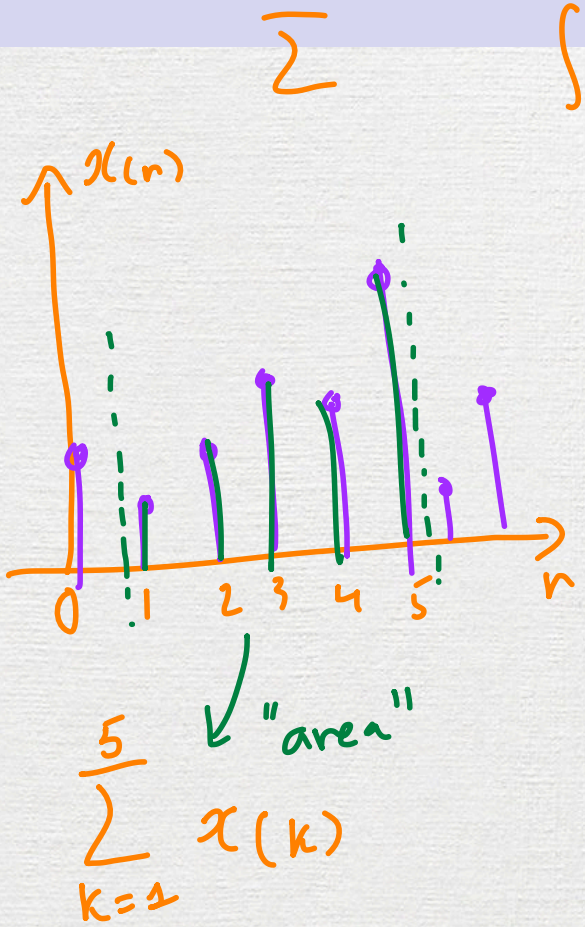
Weighted "Sum" of Shifted Impulses

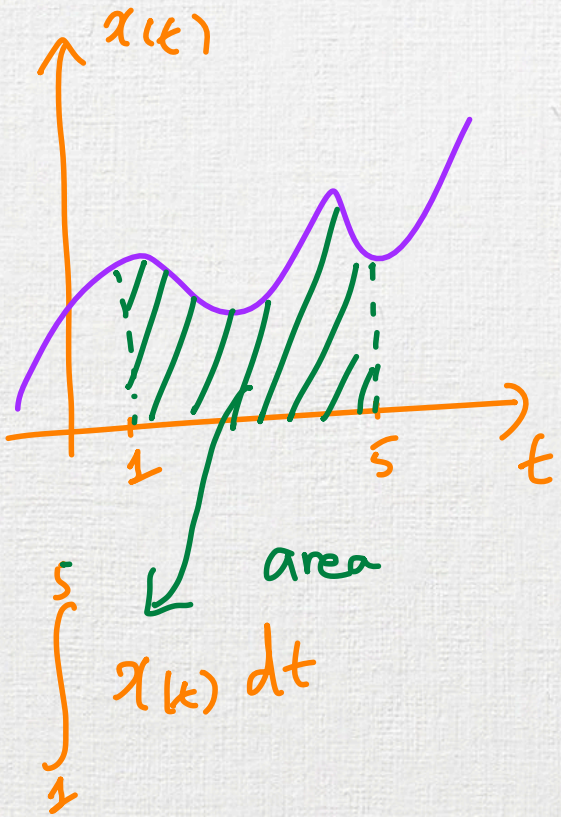
Definition

Discrete vs Continuous

Convolution Integral

Homework





Delta in Continuous-time Domain

Signals & Systems

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Objectives

Continuous Delta

Weighted "Sum" of Shifted Impulses

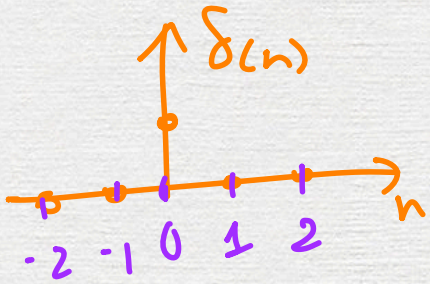
Definition

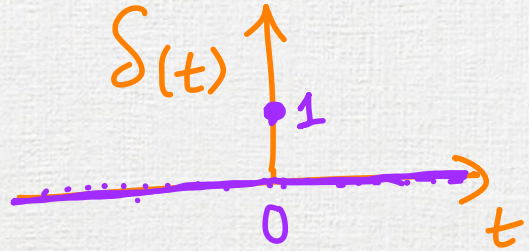
Discrete vs Continuous

Convolution Integral

Homework

Question: how should we define the continuous Delta signal?





$$\delta(t) = \begin{cases} +\infty & \text{if } t = 0 \\ 0 & \text{else} \end{cases}$$

Delta in Continuous-time Domain

Signals & Systems

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Objectives

Continuous Delta

Weighted "Sum" of Shifted Impulses

Definition

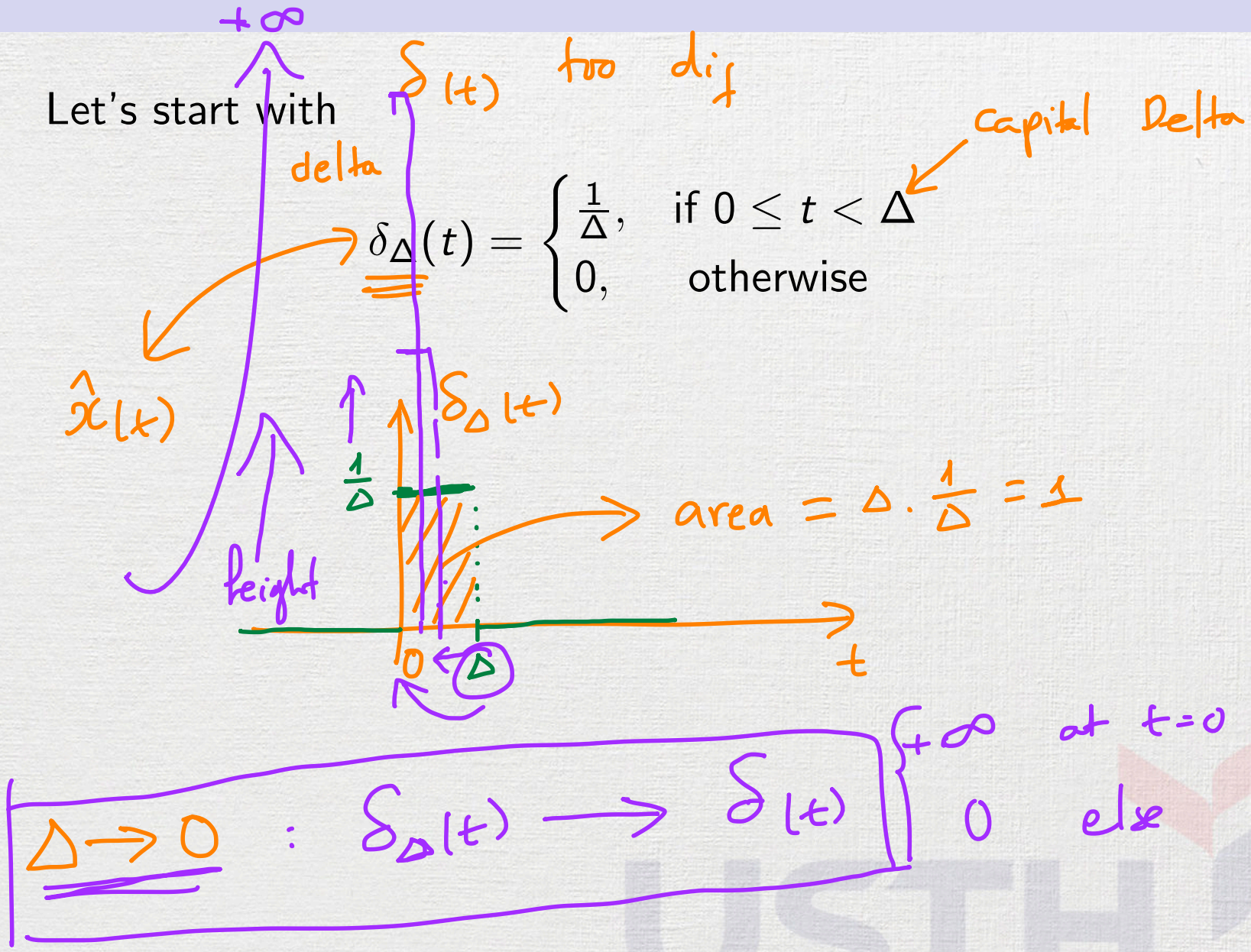
Discrete vs Continuous

Convolution Integral

Homework

Let's start with

$$\delta_\Delta(t) = \begin{cases} \frac{1}{\Delta}, & \text{if } 0 \le t < \Delta \\ 0, & \text{otherwise} \end{cases}$$



Weighted "Sum" of Shifted Impulses

Signals & Systems

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Objectives

Continuous Delta

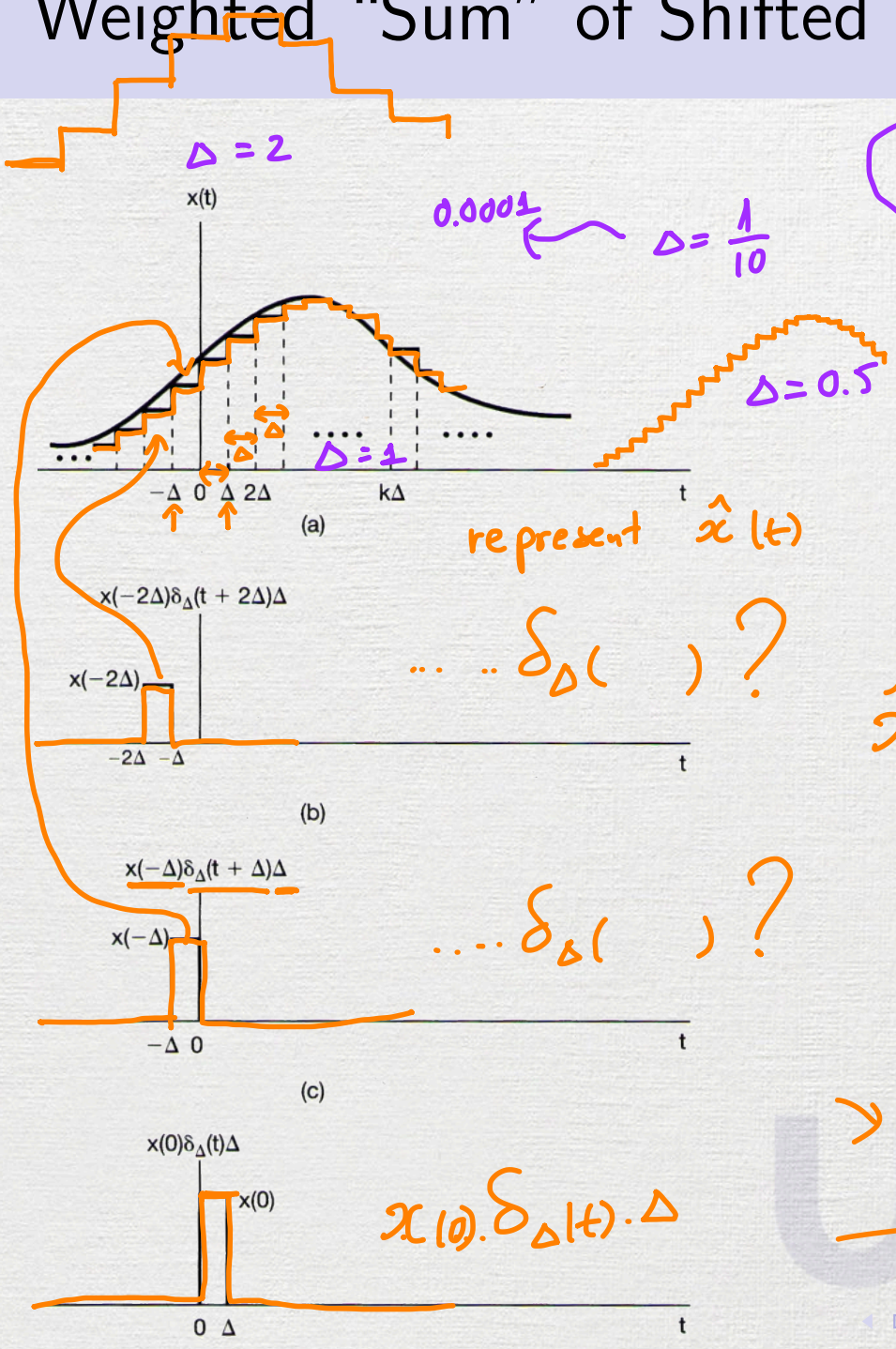
Weighted "Sum" of Shifted Impulses

Definition

Discrete vs Continuous

Convolution Integral

Homework



$\hat{x}(t) \xrightarrow{\Delta \to 0} x(t)$

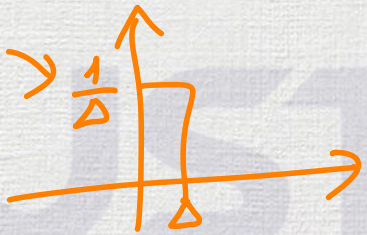
$\delta(t) \overset{?}{\rightsquigarrow} x(t)$

$\delta_\Delta(t) \rightsquigarrow \hat{x}(t)$

$\Delta \to 0$

$\hat{x}(t) = \sum_{k=-\infty}^{+\infty} x(k\Delta)\,\delta_\Delta(t - k\Delta)\,\Delta$

$x(0)\,\delta_\Delta(t)\,\Delta$



Weighted "Sum" of Shifted Impulses

Signals & Systems

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Objectives

Continuous Delta

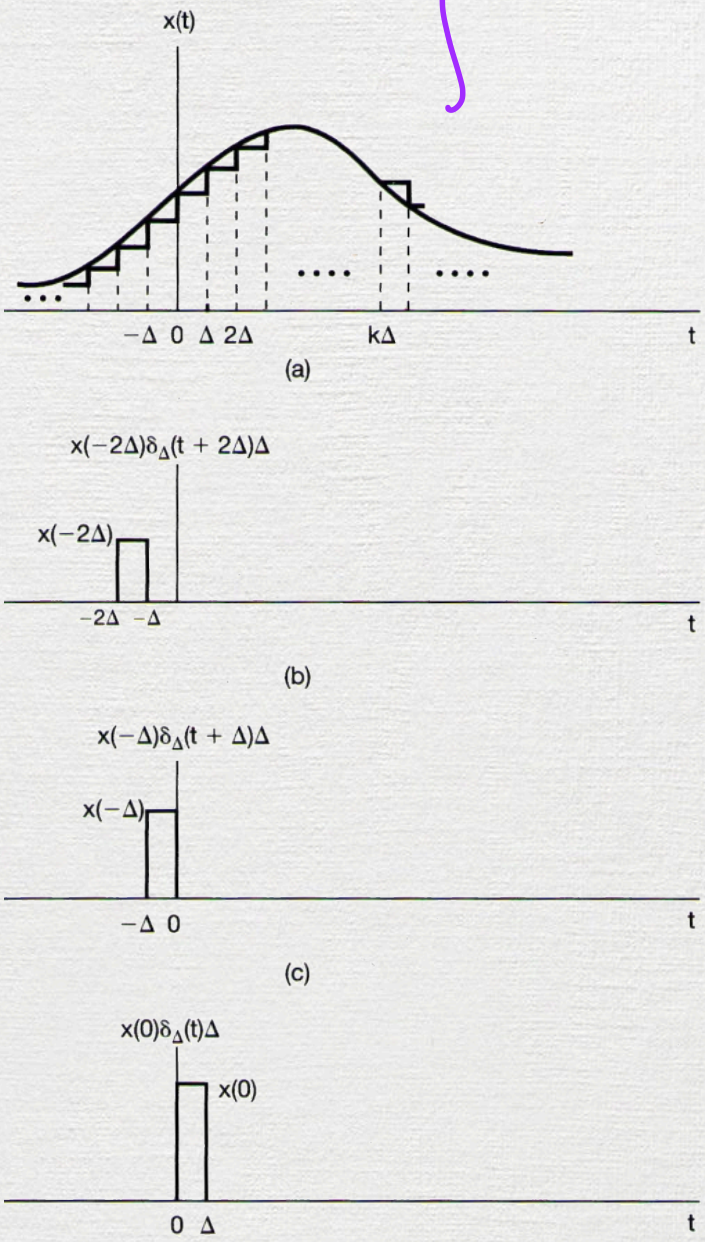
Weighted "Sum" of Shifted Impulses

Definition

Discrete vs Continuous

Convolution Integral

Homework



Approximation

$$\hat{x}(t) = \sum_{k=-\infty}^{+\infty} x(k\Delta)\delta_\Delta(t - k\Delta)\Delta$$

$$x(t) = \int_{-\infty}^{+\infty} x(\tau).\delta(t-\tau).d\tau$$

Weighted "Sum" of Shifted Impulses

Signals & Systems

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Objectives

Continuous Delta

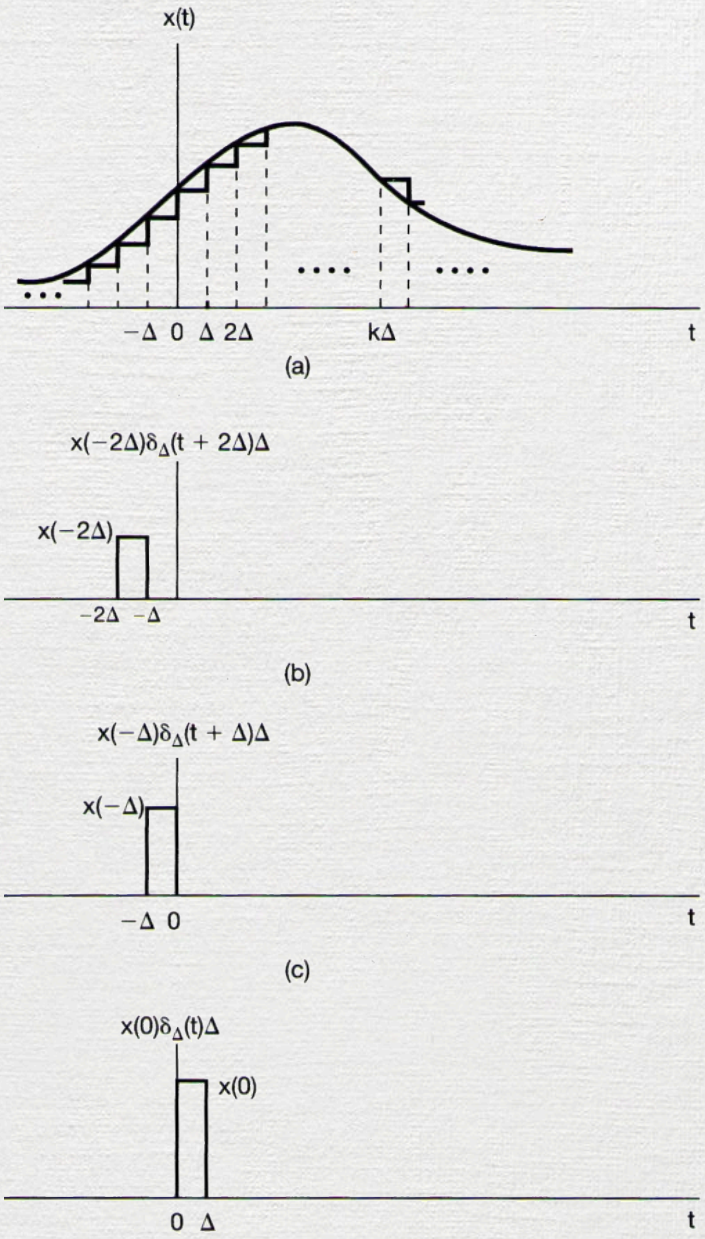
Weighted "Sum" of Shifted Impulses

Definition

Discrete vs Continuous

Convolution Integral

Homework



Approximation

$$\hat{x}(t) = \sum_{k=-\infty}^{+\infty} x(k\Delta)\delta_\Delta(t - k\Delta)\Delta$$

The smaller Δ is, the better the approximation is.

Weighted "Sum" of Shifted Impulses

Signals & Systems

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Objectives

Continuous Delta

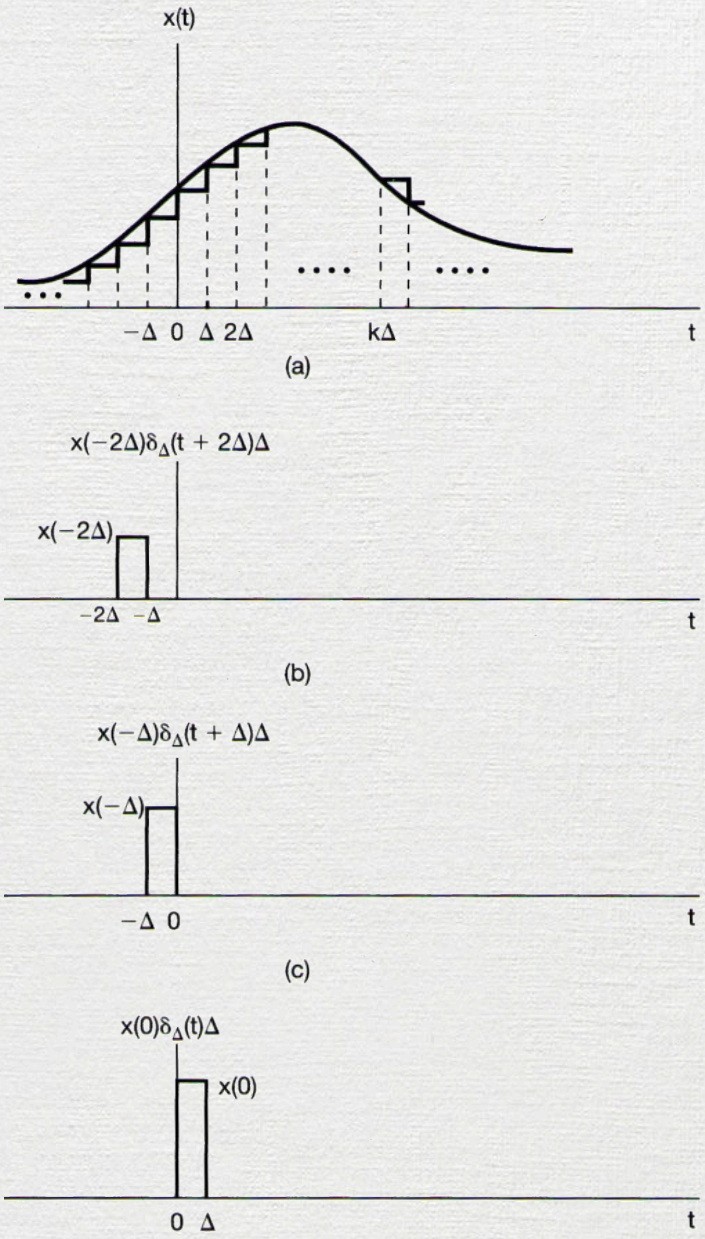
Weighted "Sum" of Shifted Impulses

Definition

Discrete vs Continuous

Convolution Integral

Homework



Approximation

$$\hat{x}(t) = \sum_{k=-\infty}^{+\infty} x(k\Delta)\delta_\Delta(t - k\Delta)\Delta$$

The smaller Δ is, the better the approximation is.

Final Result

$$x(t) = \int_{-\infty}^{+\infty} x(\tau)\delta(t - \tau)d\tau$$

Summary: Continuous-time Delta Impulse

Signals & Systems

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Objectives

Continuous Delta

Weighted "Sum" of Shifted Impulses

Definition

Discrete vs Continuous

Convolution Integral

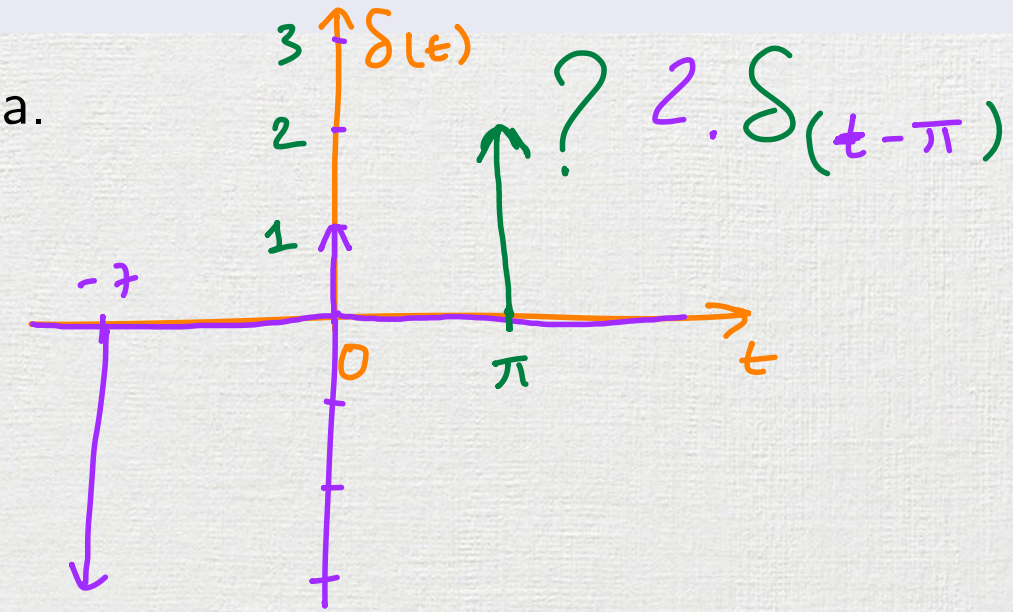
Homework

Delta Signal

$$\delta(t) = \begin{cases} +\infty & \text{if } t = 0 \\ 0 & \text{if } t \neq 0. \end{cases}$$

$2\,\delta(t)$

Plot the Delta.



Summary: Continuous-time Delta Impulse

Signals & Systems

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Objectives

Continuous Delta

Weighted "Sum" of Shifted Impulses

Definition

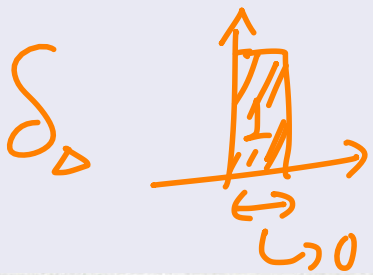
Discrete vs Continuous

Convolution Integral

Homework

Delta Signal

$$\delta(t) = \begin{cases} +\infty & \text{if } t = 0 \\ 0 & \text{if } t \neq 0. \end{cases}$$



Properties:

1. $$\int_{-\infty}^{+\infty} \delta(t)dt = 1$$

Summary: Continuous-time Delta Impulse

Signals & Systems

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Objectives

Continuous Delta

Weighted "Sum" of Shifted Impulses

Definition

Discrete vs Continuous

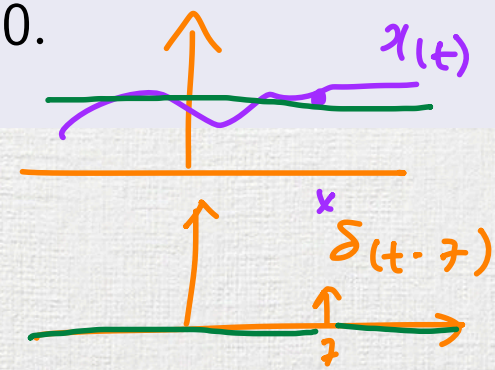
Convolution Integral

Homework

Delta Signal

$$\delta(t) = \begin{cases} +\infty & \text{if } t = 0 \\ 0 & \text{if } t \neq 0. \end{cases}$$

$0 \cdot a = 0$



Properties:

1

$$\int_{-\infty}^{+\infty} \delta(t)dt = 1$$

2

$$\int_{-\infty}^{+\infty} x(t)\delta(t-\tau)dt = x(\tau)$$

verify?

$x(\tau) \cdot \int_{-\infty}^{+\infty} \delta(t-\tau)\,dt$

Summary: Continuous-time Delta Impulse

Signals & Systems

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Continuous Delta

Weighted "Sum" of Shifted Impulses

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Discrete vs Continuous

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Homework

Delta Signal

$$\delta(t) = \begin{cases} +\infty & \text{if } t = 0 \\ 0 & \text{if } t \neq 0. \end{cases}$$

Properties:

1. $$\int_{-\infty}^{+\infty} \delta(t)dt = 1$$

2. $$\int_{-\infty}^{+\infty} x(t)\delta(t-\tau)dt = x(\tau)$$

Justify the second property!

Discrete Delta vs Continuous Delta

Signals & Systems

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Objectives

Continuous Delta

Weighted "Sum" of Shifted Impulses

Definition

Discrete vs Continuous

Convolution Integral

Homework

$$\delta[n] = \begin{cases} 1 & \text{if } n = 0 \\ 0 & \text{if } n \neq 0. \end{cases}$$

$$\delta(t) = \begin{cases} +\infty & \text{if } t = 0 \\ 0 & \text{if } t \neq 0. \end{cases}$$

Discrete Delta vs Continuous Delta

Signals & Systems

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Continuous Delta

Weighted "Sum" of Shifted Impulses

Definition

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Homework

$$\delta[n] = \begin{cases} 1 & \text{if } n = 0 \\ 0 & \text{if } n \neq 0. \end{cases}$$

$$\sum_{n=-\infty}^{+\infty} \delta[n] = 1$$

$$\delta(t) = \begin{cases} +\infty & \text{if } t = 0 \\ 0 & \text{if } t \neq 0. \end{cases}$$

$$\int_{-\infty}^{+\infty} \delta(t)dt = 1$$

Discrete Delta vs Continuous Delta

Signals & Systems

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Objectives

Continuous Delta

Weighted "Sum" of Shifted Impulses

Definition

Discrete vs Continuous

Convolution Integral

Homework

$$\delta[n] = \begin{cases} 1 & \text{if } n = 0 \\ 0 & \text{if } n \neq 0. \end{cases}$$

$$\sum_{n=-\infty}^{+\infty} \delta[n] = 1$$

$$\sum_{n=-\infty}^{+\infty} x[n]\delta[n - n_0] = x[n_0]$$

$$\delta(t) = \begin{cases} +\infty & \text{if } t = 0 \\ 0 & \text{if } t \neq 0. \end{cases}$$

$$\int_{-\infty}^{+\infty} \delta(t)dt = 1$$

$$\int_{-\infty}^{+\infty} x(t)\delta(t - \tau)dt = x(\tau)$$

Discrete Delta vs Continuous Delta

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Continuous Delta

Weighted "Sum" of Shifted Impulses

Definition

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Homework

$$\delta[n] = \begin{cases} 1 & \text{if } n = 0 \\ 0 & \text{if } n \neq 0. \end{cases}$$

$$\sum_{n=-\infty}^{+\infty} \delta[n] = 1$$

$$\sum_{n=-\infty}^{+\infty} x[n]\delta[n - n_0] = x[n_0]$$

$$x[n] = \sum_{k=-\infty}^{+\infty} x[k]\delta[n - k]$$

$$\delta(t) = \begin{cases} +\infty & \text{if } t = 0 \\ 0 & \text{if } t \neq 0. \end{cases}$$

$$\int_{-\infty}^{+\infty} \delta(t)dt = 1$$

$$\int_{-\infty}^{+\infty} x(t)\delta(t - \tau)dt = x(\tau)$$

$$x(t) = \int_{-\infty}^{+\infty} x(\tau)\delta(t - \tau)d\tau$$

$\delta(t)$ and $u(t)$

Signals & Systems

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Objectives

Continuous Delta

Weighted "Sum" of Shifted Impulses

Definition

Discrete vs Continuous

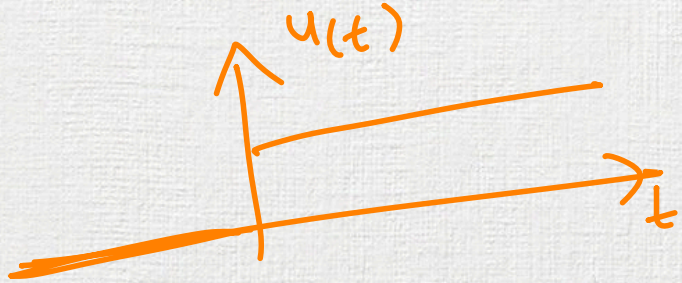
Convolution Integral

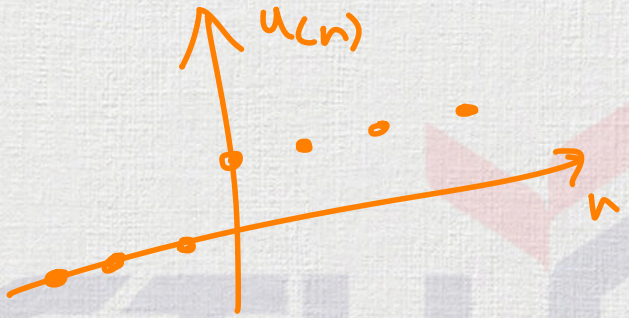
Homework

$$\begin{cases} +\infty \\ 0 \end{cases}$$

$$u(t) = \int_{-\infty}^{t} \delta(\tau)\, d\tau$$

$$u(t) = \begin{cases} 1 & \text{if } t \geq 0 \\ 0 & \text{else} \end{cases}$$





1 Lesson Objectives

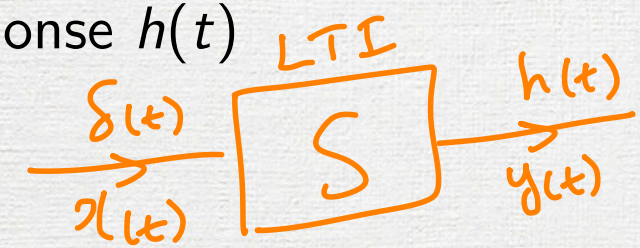
2 Continuous-time Delta

3 Convolution Integral
- Definition
- Examples

4 Homework

Convolution Integral of a LTI system

Impulse response $h(t)$





$$x(t) = \int_{-\infty}^{+\infty} x(\tau).\delta(t-\tau).d\tau$$

$$y(t) = x(t) * h(t) = \int_{-\infty}^{+\infty} x(\tau).h(t-\tau).d\tau$$

Convolution Integral of a LTI system

Signals & Systems

TRAN Hoang Tung

Objectives

Continuous Delta

Convolution Integral

Definition

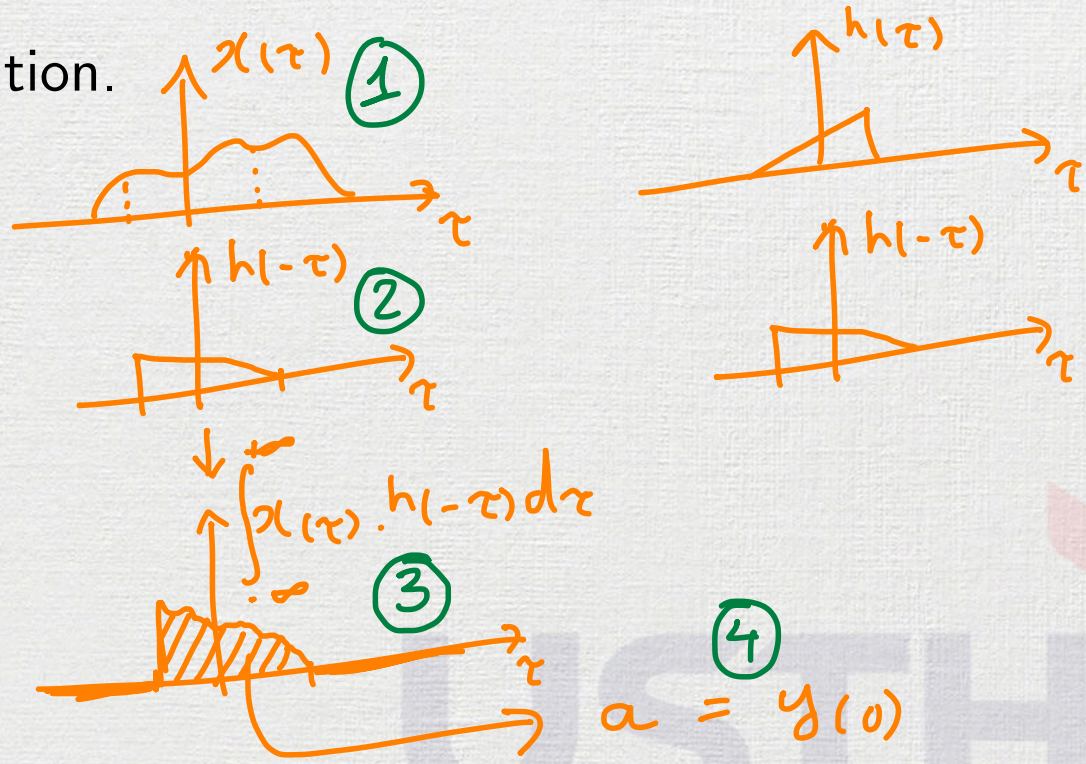
Examples

Homework

Definition

$$y(t) = \int_{-\infty}^{+\infty} x(\tau)h(t-\tau)d\tau$$

Visual explanation.



Signals & Systems

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Objectives

Continuous Delta

Convolution Integral

Definition

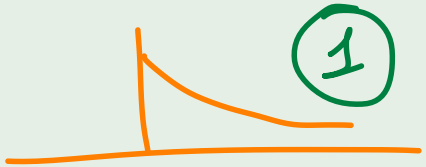
Examples

Homework

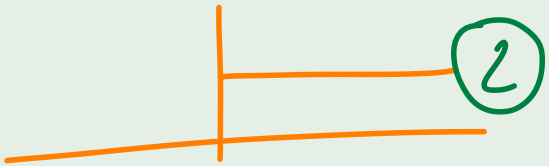
Example 1

Consider the convolution of the following two signals:

$$x(t) = e^{-2t}u(t)$$



$$h(t) = u(t)$$







Convolution $y(t) = \int_{-\infty}^{+\infty} x(\tau)h(t-\tau)d\tau$

Signals & Systems

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Objectives

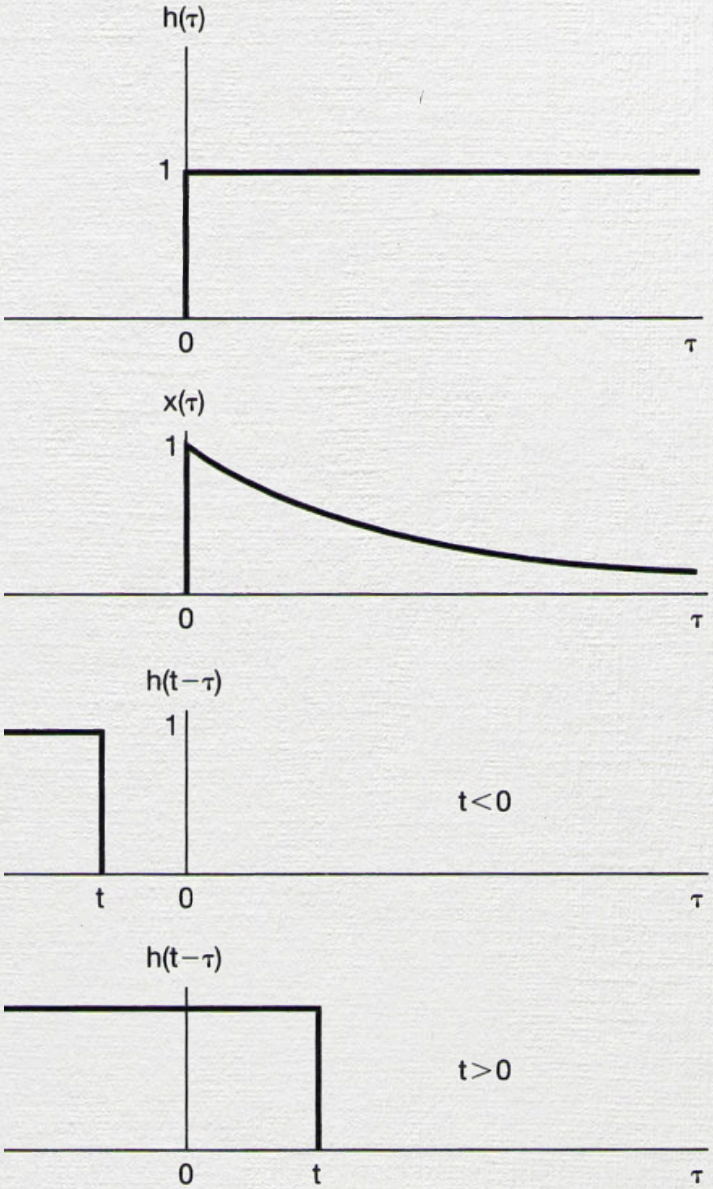
Continuous Delta

Convolution Integral

Definition

Examples

Homework



Signals & Systems

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Objectives

Continuous Delta

Convolution Integral

Definition

Examples

Homework

Example 2

Consider the convolution of the following two signals:

$$x(t) = \begin{cases} 1, & 0 < t < T \\ 0, & otherwise \end{cases}$$

$$h(t) = \begin{cases} t, & 0 < t < 2T \\ 0, & otherwise \end{cases}$$

Convolution $y(t) = \int_{-\infty}^{+\infty} x(\tau)h(t-\tau)d\tau$

Signals & Systems

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Exercises

Signals & Systems

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Examples

Homework

Determine and sketch the convolution of the following two signals:

$$x(t) = \begin{cases} t+1, & 0 \le t \le 1 \\ 2-t, & 1 < t \le 2 \\ 0, & \text{elsewhere} \end{cases},$$

$$h(t) = \delta(t+2) + 2\delta(t+1).$$

Exercises

Signals & Systems

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Objectives

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Definition

Examples

Homework

Suppose that

$$x(t) = \begin{cases} 1, & 0 \leq t \leq 1 \\ 0, & \text{elsewhere} \end{cases}$$

and $h(t) = x(t/\alpha)$, where $0 < \alpha \leq 1$.

Determine and sketch $y(t) = x(t) * h(t)$.

1 Lesson Objectives

2 Continuous-time Delta

3 Convolution Integral

4 Homework

Homework

Signals & Systems

TRAN Hoang Tung

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Continuous Delta

Convolution Integral

Homework

Continuous LTI Systems Exercises

2.8, 2.9, 2.10, 2.22, 2.23